ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 1ST YEAR

DISCRETE MATHEMATICS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BSc Computer Science, 1st year, semester 1 final examination: logic and predicates, sets, relations and functions, methods of proof and mathematical induction, counting, and graph theory.

Written by Academic Central for practice, following first-year Computer Science and Mathematics syllabuses at South African universities (discrete mathematics for computer science). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SIX questions. Answer ALL the questions.
2. Show ALL your working. Marks are given for correct steps, not only for final answers.
3. Name the law or rule you use where a question asks for it.
4. Non-programmable calculators may be used.
5. Number your answers exactly as the questions are numbered.

QUESTION 1: LOGIC

1.1 Let p be "It is raining" and q be "I take a taxi". Write each statement in symbols:

1.1.1 If it is raining, then I take a taxi. (1)

1.1.2 I take a taxi only if it is raining. (1)

1.1.3 It is not raining and I do not take a taxi. (1)

1.2 Construct a truth table for $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$. Is the statement a tautology? Give a reason. (5)

1.3 Use the laws of logic (NOT a truth table) to show that $\neg(p \vee \neg q) \vee (\neg p \wedge \neg q) \equiv \neg p$. Name the law used in each step. (4)

1.4 Write down the converse, the inverse and the contrapositive of: "If n is even, then n^2 is even." (3)

1.5 The domain is the set of integers $\mathbb{Z}$. State whether each statement is TRUE or FALSE and give a reason:

1.5.1 $\forall x\,(x^2 \geq 0)$ (1)

1.5.2 $\exists x\,(x^2 = 2)$ (1)

1.5.3 $\forall x\,\exists y\,(x + y = 0)$ (1)

1.6 Write down the negation of $\forall x\,\exists y\,(x < y)$ so that no negation symbol stands in front of a quantifier. (2)

[20]

QUESTION 2: SETS

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{2, 4, 6, 8, 10\}$ and $C = \{3, 6, 9\}$.

2.1 List the elements of:

2.1.1 $A \cap B$ (1)

2.1.2 $A \cup C$ (1)

2.1.3 $B - A$ (1)

2.1.4 $(A \cup B)'$, the complement of $A \cup B$ (1)

2.1.5 How many elements does $A \times C$ have? (1)

2.2 How many elements does the power set P(C) have? List the subsets of C that have exactly two elements. (2)

2.3 Prove, using an element argument, that for any sets A, B and C: $A - (B \cup C) = (A - B) \cap (A - C)$. (4)

2.4 Of 120 first-year students, 70 take Mathematics, 50 take Physics and 30 take both.

2.4.1 How many students take neither subject? Show your working. (2)

2.4.2 How many take Mathematics only, and how many take Physics only? (2)

[15]

QUESTION 3: RELATIONS AND FUNCTIONS

Let A = {1, 2, 3, 4} and let R = {(1,1), (2,2), (3,3), (4,4), (1,2), (2,1), (3,4), (4,3)} be a relation on A.

3.1 Is R reflexive, symmetric, antisymmetric and transitive? Give a reason for each answer. (4)

3.2 Is R an equivalence relation? If it is, write down its equivalence classes. (3)

3.3 Let $f: \mathbb{R} \to \mathbb{R}$ with $f(x) = 3x - 5$, and $g: \mathbb{R} \to \mathbb{R}$ with $g(x) = x^2 + 1$.

3.3.1 Calculate $(g \circ f)(2)$. (1)

3.3.2 Find $(f \circ g)(x)$ in its simplest form. (2)

3.3.3 Show that f is one-to-one and onto, and find $f^{-1}(x)$. (3)

3.4 Is g one-to-one? Justify your answer. (2)

[15]

QUESTION 4: METHODS OF PROOF

4.1 Give a direct proof that if n is an odd integer, then n^2 is odd. (4)

4.2 Prove by contraposition: for any integer n, if $3n + 2$ is odd, then n is odd. (4)

4.3 Use mathematical induction to prove that for every integer $n \geq 1$: (7)
$1 + 3 + 5 + \ldots + (2n - 1) = n^2$

[15]

QUESTION 5: COUNTING

5.1 A student login code has 2 letters (A–Z) followed by 4 digits (0–9).

5.1.1 How many different codes are possible? (2)

5.1.2 How many codes are possible if the 4 digits must all be different? (2)

5.2 A committee of 4 is chosen from 7 women and 5 men.

5.2.1 How many different committees are possible? (2)

5.2.2 How many committees have exactly 2 women? (2)

5.2.3 How many committees have at least one man? (2)

5.3 How many different arrangements of the letters of the word LETTER are there? (2)

5.4 Use the pigeonhole principle to explain why, in any group of 13 people, at least two were born in the same month. (2)

5.5 Find the coefficient of x^3 in the expansion of $(1 + x)^7$. (1)

[15]

QUESTION 6: GRAPH THEORY

Answer 6.1 to 6.5 using the graph G in FIGURE 1.

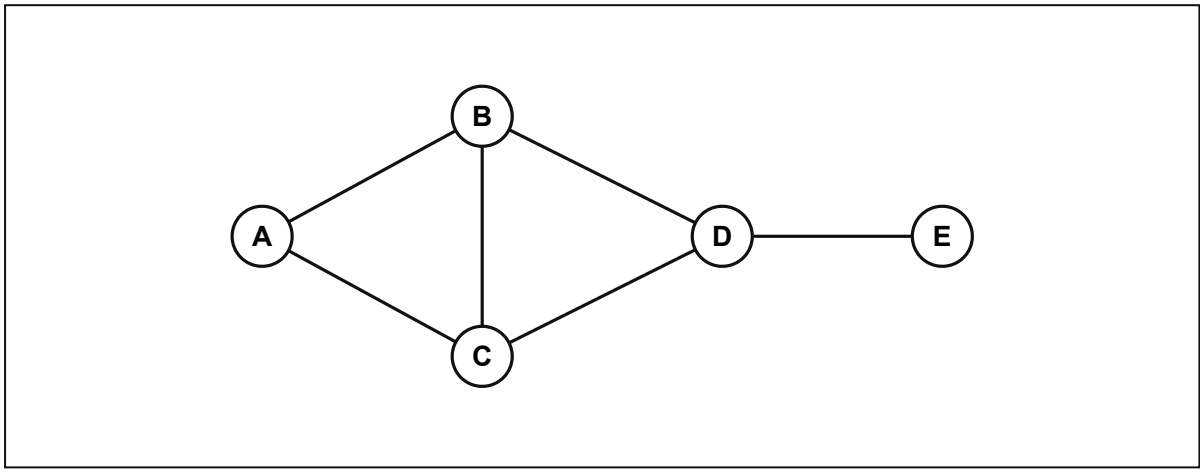


FIGURE 1: Graph G

6.1 How many vertices and how many edges does G have? (2)

6.2 Give the degree of each vertex, and show that the handshaking lemma holds for G. (3)

6.3 Does G have an Euler circuit? Does it have an Euler path? Justify your answers. (3)

6.4 Write down the adjacency matrix of G (vertices in the order A, B, C, D, E). (3)

6.5 Is G a tree? Give a reason. (2)

6.6 Use Dijkstra's algorithm on the weighted graph H in FIGURE 2 to find the shortest path from S to T and its length. Show the table of distances at each step. (5)

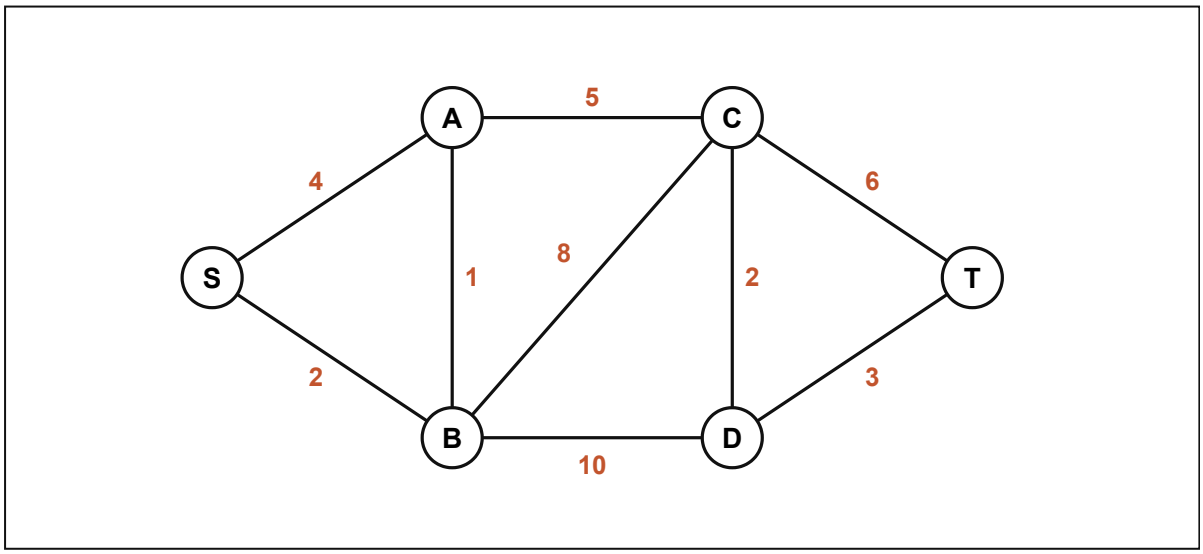


FIGURE 2: Weighted graph H

6.7 How many edges does the complete graph K_6 have? (2)

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TOTAL: 100

END OF QUESTION PAPER